

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457370.67968 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1435 +/- 0.0052
Transit depth (Rp/Rs)^2	2.06 +/- 0.15 [%]
Orbital Inclination (inc)	81.1 +/- 1.4
Airmass coefficient 1 (a1)	1.1429 +/- 0.0098
Airmass coefficient 2 (a2)	-0.1159 +/- 0.0011
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.38
Optimal Annulus	15.14
Transit Duration (day)	0.1227 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1435 ± 0.0052 vs 0.1178 (deviation 3.3σ)
Duration fit vs published	0.1227 d vs 0.125 d (deviation 0.41σ)
Mid-time O–C	-0.7 min
Depth SNR	18.4
Residual scatter	0.82 %

Light curve

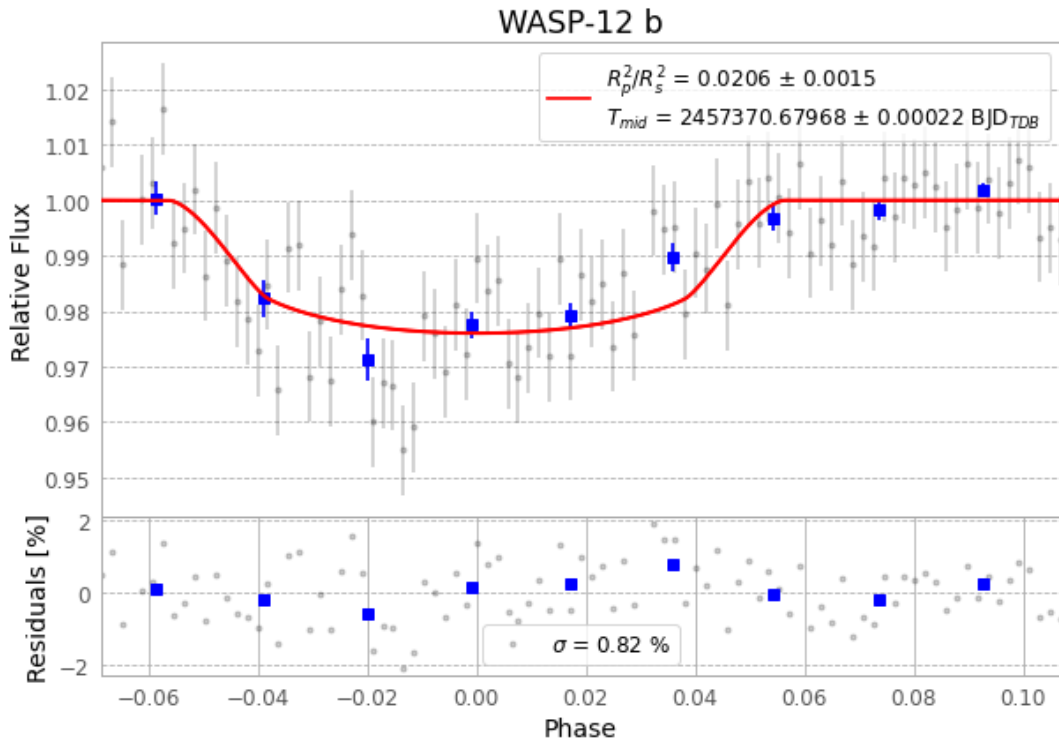


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-13.png (SHA-256 b65c840512c2b879...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 143], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 faf2bdef4df4bca5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

93 FITS frames (61 MB), WASP-12151214022112.FITS → WASP-12151214065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151214.log (cb644d5e04ad...), FinalLightCurve_WASP-12 b_2015-12-13.png (b65c840512c2...), FinalLightCurve_WASP-12 b_2015-12-13.csv (0e4dffd0887c...)

Compute resources

Wall time 137.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 22:51Z.